

Heart Disease Prediction using Machine Learning

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1. Introduction

This project focuses on analyzing a dataset related to heart disease and building a machine learning model to predict the presence of heart disease based on clinical and demographic features. Early detection of heart disease can help in timely treatment and prevention of severe health conditions.

The dataset contains various attributes such as age, sex, cholesterol level, blood pressure, and other medical indicators that influence heart health.

2. Data Preprocessing

- Checked for missing values using `df.isnull().sum()`
- No missing values were found in the dataset.
- No categorical encoding was required as data is already numeric.
- Features and target variable were separated:
 - Features (X): All columns except target
 - Target (y): Heart disease presence (0 or 1)
- Data was prepared for training and testing.

```
[1]: import pandas as pd

# Load dataset
df = pd.read_csv("heart.csv")

# Display first 5 rows
print("First 5 rows:")
print(df.head())

# Basic info
print("\nDataset Info:")
print(df.info())

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

First 5 rows:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	\
0	52	1	0	125	212	0	1	168	0	1.0	2	
1	53	1	0	140	203	1	0	155	1	3.1	0	
2	70	1	0	145	174	0	1	125	1	2.6	0	
3	61	1	0	148	203	0	1	161	0	0.0	2	
4	62	0	0	138	294	1	1	106	0	1.9	1	

	ca	thal	target
0	2	3	0
1	0	3	0
2	0	3	0
3	1	3	0
4	3	2	0

```
[4]: # Check missing values
print(df.isnull().sum())

# Since no missing values, directly split data
X = df.drop('target', axis=1)
y = df['target']

age      0
sex      0
cp       0
trestbps 0
chol     0
fbs      0
restecg  0
thalach  0
exang    0
oldpeak  0
slope    0
ca       0
thal     0
target   0
dtype: int64
```

3. Exploratory Data Analysis (EDA)

- Examined dataset structure using:
 - df.head()
 - df.info()
 - df.describe()
- Observed distribution of features such as age, cholesterol, and heart rate.

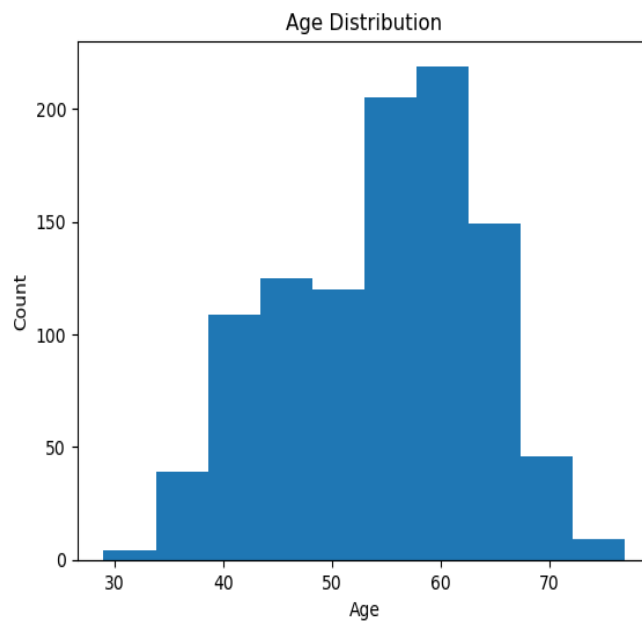
- Correlation matrix was analyzed to understand relationships between features.
- Identified which features are strongly related to heart disease.

```
[10]: import matplotlib.pyplot as plt

# Histogram (Age)
plt.hist(df['age'])
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()

# Boxplot (Cholesterol)
plt.boxplot(df['chol'])
plt.title("Cholesterol Distribution")
plt.show()

# Correlation Matrix (print)
print("Correlation Matrix")
print(df.corr())
```





Correlation Matrix

	age	sex	cp	trestbps	chol	fbs	\
age	1.000000	-0.103240	-0.071966	0.271121	0.219823	0.121243	
sex	-0.103240	1.000000	-0.041119	-0.078974	-0.198258	0.027200	
cp	-0.071966	-0.041119	1.000000	0.038177	-0.081641	0.079294	
trestbps	0.271121	-0.078974	0.038177	1.000000	0.127977	0.181767	
chol	0.219823	-0.198258	-0.081641	0.127977	1.000000	0.026917	
fbs	0.121243	0.027200	0.079294	0.181767	0.026917	1.000000	
restecg	-0.132696	-0.055117	0.043581	-0.123794	-0.147410	-0.104051	
thalach	-0.390227	-0.049365	0.306839	-0.039264	-0.021772	-0.008866	
exang	0.088163	0.139157	-0.401513	0.061197	0.067382	0.049261	
oldpeak	0.208137	0.084687	-0.174733	0.187434	0.064880	0.010859	
slope	-0.169105	-0.026666	0.131633	-0.120445	-0.014248	-0.061902	
ca	0.271551	0.111729	-0.176206	0.104554	0.074259	0.137156	
thal	0.072297	0.198424	-0.163341	0.059276	0.100244	-0.042177	
target	-0.229324	-0.279501	0.434854	-0.138772	-0.099966	-0.041164	

	restecg	thalach	exang	oldpeak	slope	ca	\
age	-0.132696	-0.390227	0.088163	0.208137	-0.169105	0.271551	
sex	-0.055117	-0.049365	0.139157	0.084687	-0.026666	0.111729	
cp	0.043581	0.306839	-0.401513	-0.174733	0.131633	-0.176206	
trestbps	-0.123794	-0.039264	0.061197	0.187434	-0.120445	0.104554	
chol	-0.147410	-0.021772	0.067382	0.064880	-0.014248	0.074259	
fbs	-0.104051	-0.008866	0.049261	0.010859	-0.061902	0.137156	
restecg	1.000000	0.048411	-0.065606	-0.050114	0.086086	-0.078072	
thalach	0.048411	1.000000	-0.380281	-0.349796	0.395308	-0.207888	
exang	-0.065606	-0.380281	1.000000	0.310844	-0.267335	0.107849	
oldpeak	-0.050114	-0.349796	0.310844	1.000000	-0.575189	0.221816	
slope	0.086086	0.395308	-0.267335	-0.575189	1.000000	-0.073440	
ca	-0.078072	-0.207888	0.107849	0.221816	-0.073440	1.000000	
thal	-0.020504	-0.098068	0.197201	0.202672	-0.094090	0.149014	
target	0.134468	0.422895	-0.438029	-0.438441	0.345512	-0.382085	

	thal	target
age	0.072297	-0.229324
sex	0.198424	-0.279501
cp	-0.163341	0.434854
trestbps	0.059276	-0.138772
chol	0.100244	-0.099966
fbs	-0.042177	-0.041164
restecg	-0.020504	0.134468
thalach	-0.098068	0.422895
exang	0.197201	-0.438029
oldpeak	0.202672	-0.438441
slope	-0.094090	0.345512
ca	0.149014	-0.382085
thal	1.000000	-0.337838
target	-0.337838	1.000000

4. Model Selection & Training

- Logistic Regression model was used for classification.
- The model was trained using training data.
- Reason for choosing Logistic Regression:

- Simple and effective for binary classification.
- Works well with structured datasets.

```
[11]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
print("Training size:", len(X_train))
print("Testing size:", len(X_test))
```

```
Training size: 820
Testing size: 205
```

```
[14]: from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
print("Model trained successfully")
```

```
Model trained successfully
```

5. Evaluation & Results

The performance of the machine learning model was evaluated using standard classification metrics to measure its effectiveness in predicting the presence of heart disease.

The dataset was divided into training and testing sets, and the model was trained using the training data. The evaluation was then performed on the test data to ensure unbiased results.

The following metrics were used:

- **Accuracy:** Measures the overall correctness of the model. It indicates the percentage of correctly predicted instances out of the total observations.

- **Precision:** Represents the proportion of correctly predicted positive cases out of all predicted positive cases.
- **Recall (Sensitivity):** Measures the model's ability to correctly identify actual positive cases (patients with heart disease).
- **F1-Score:** The harmonic mean of precision and recall.

A **confusion matrix** was also generated to visualize the performance of the model. It shows:

- True Positives (correctly predicted heart disease cases)
- True Negatives (correctly predicted non-disease cases)
- False Positives (incorrectly predicted disease)
- False Negatives (missed disease cases)

The results indicate that the model performs well in distinguishing between patients with and without heart disease. However, some misclassifications were observed, which suggests that there is still scope for improvement.

```
[15]: from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

```
print("\nClassification Report:")
```

```
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.7951219512195122
```

```
Confusion Matrix:
```

```
[[73 29]
```

```
 [13 90]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.85	0.72	0.78	102
1	0.76	0.87	0.81	103
accuracy			0.80	205
macro avg	0.80	0.79	0.79	205
weighted avg	0.80	0.80	0.79	205

6. Conclusion

- The machine learning model successfully predicts heart disease based on input features
- Features like age, cholesterol, and maximum heart rate play an important role
- The model can be improved further using advanced algorithms like Random Forest or SVM
- This approach can help in early diagnosis and medical decision-making

Google Drive Link:

https://drive.google.com/drive/folders/1XXVZ2njHe0xySKw-UgKK_yJ8L0wJBIKW?usp=drive_link

